Introduction to Deep Learning

Session 9: Convolutional Neural Networks

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Lecture Overview

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1. Why Not MLPs for Images?
2. Convolution Operation
3. Deep CNNs
4. Pooling
5. CNN Architecture
6. Classic Architectures
7. Modern Details
8. Summary

Images in Computers

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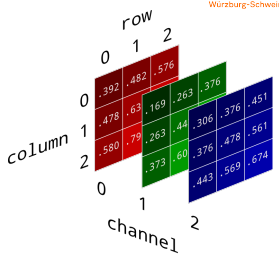

Image: 3D tensor of shape $C \times H \times W$

- Each entry: pixel intensity, integer 0–255 or float 0–1,
- 2^8 possible values $\sim$ 8 bits
- **Grayscale** ($C = 1$): single channel, 0 = black, 255 = white
- **RGB** ($C = 3$): red, green, blue — standard cameras, screens

Other formats:

- RGBA ($C = 4$): RGB + α transparency
- CMYK ($C = 4$): print format
- Medical/hyperspectral: many channels, often 16-bit depth

DL mostly **RGB and grayscale** — other formats typically converted before use

Images as Data

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Image sizes — from benchmarks to real world:

Source	Size	Channels	Pixels
MNIST	28×28	1	784
ImageNet	224×224	3	150 528
Phone camera	4000×3000	3	36 000 000
Full HD screen	1920×1080	3	6 220 800
Medical (CT slice)	512×512	1	262 144

MLP approach: flatten image to vector $\mathbf{x} \in \mathbb{R}^{H \cdot W \cdot C}$, apply fully connected layers

ImageNet: first hidden from $\mathbb{R}^{150\,528} \to \mathbb{R}^{1024} \Rightarrow \approx 154$M parameters — one layer only!

How to handle high-dimensional data efficiently?

How to Look at Images

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Human vision:

- Looks for local patterns — edges, corners, textures
- Same pattern recognised anywhere in the visual field
- Simple features combine into complex ones

Technical equivalent:

- **Local connectivity** — connect to small patches only
- **Parameter sharing** — same filter at every location
- **Hierarchical features** — edges → shapes → objects

Three key properties: **locality**, **translation invariance**, **parameter efficiency**

These define the **convolutional layer**

Convolution Operation

Convolution — The Operation

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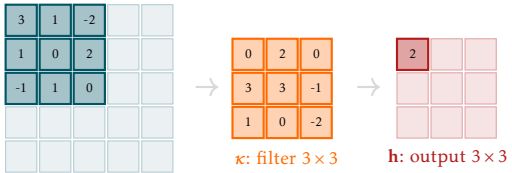
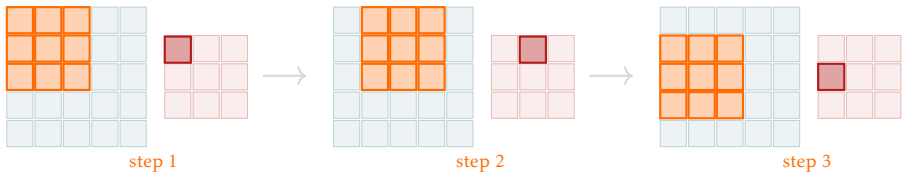
Single step: dot product of local patch and filter → one output value



Full operation: same filter slides across input — each position produces one output value



All 3 requirements: local connectivity, parameter sharing, translation invariance

1D Convolution

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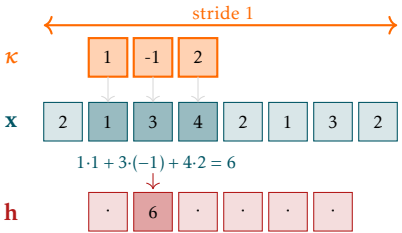
Filter $\kappa \in \mathbb{R}^k$ slides over signal $\mathbf{x} \in \mathbb{R}^n$ — at each position i, dot product with local patch



$$(\mathbf{x} * \boldsymbol{\kappa})[i] = \sum_{j=0}^{k-1} \mathbf{x}[i+j]\,\boldsymbol{\kappa}[j]$$

- i — output position; j — index within filter
- Dot product of filter with local patch

Example: at position $i = 1$:

$$1\cdot 1 + 3\cdot(-1) + 4\cdot 2 = 6$$

2D Convolution

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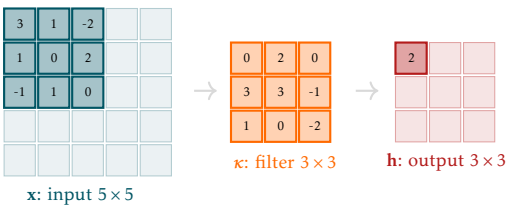
Extend to 2D: filter $\kappa \in \mathbb{R}^{k \times k}$ slides over image $\mathbf{X} \in \mathbb{R}^{H \times W}$:

$$(\mathbf{X} * \kappa)[i, j] = \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[i+p, j+q]\, \kappa[p, q]$$

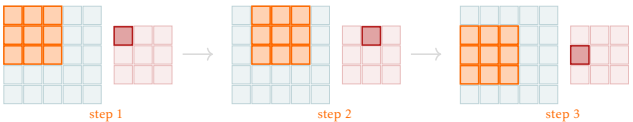


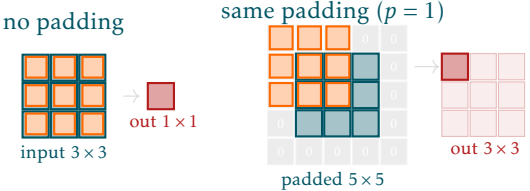
- (i, j) — output position
- (p, q) — index within filter
- Two sums: over rows and columns of filter

Output size: $(H - k + 1) \times (W - k + 1)$

Padding

Problem: convolution shrinks spatial dimensions — each layer loses $k-1$ pixels





Valid padding ($p = 0$)

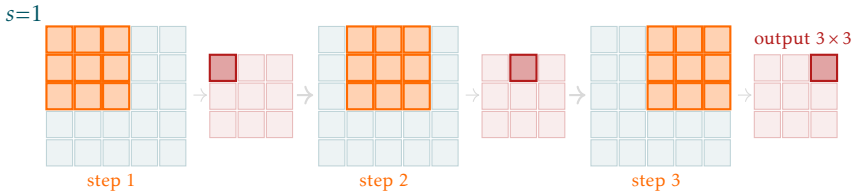
- No zeros added
- Output shrinks: $(H-k+1) \times (W-k+1)$
- Spatial information lost at borders

Same padding ($p = (k-1)/2$)

- Zeros added around border
- Output same size as input
- Standard choice in most architectures

Stride

Stride s: filter κ moves s pixels at each step instead of 1



Stride 1: κ moves one step a time

Stride 2: κ skips one - output halved

General output size: $\left\lfloor \frac{H + 2p - k}{s} \right\rfloor + 1$

Multiple Input Channels

RGB input: C_{in} channels — filter must have same depth C_{in}

Operation: filter has C_{in} slices — each slice convolves with one input channel

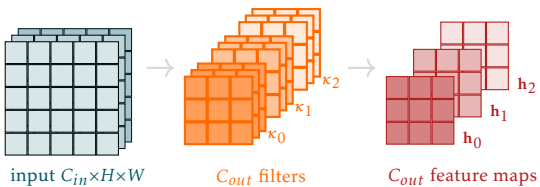
$$(\mathbf{X} * \boldsymbol{\kappa})[i,j] = \sum_{c=0}^{C_{in}-1} \underbrace{\sum_{p=0}^{k-1}\sum_{q=0}^{k-1} \mathbf{X}[c, i+p, j+q]\, \boldsymbol{\kappa}[c,p,q]}_{\text{conv with channel } c}$$

- Inner double sum: convolution of filter slice c with input channel c
- Outer sum over c: results summed across all channels
- One filter → one output feature map

Multiple Output Channels

C_{out} **filters:** each filter independently applied to the full input volume

$$\mathbf{h}_f[i,j] = \sum_{c=0}^{C_{in}-1} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[c, i+p, j+q]\, \boldsymbol{\kappa}_f[c,p,q] \quad f = 0,\ldots,C_{out}-1$$



- Each filter $\boldsymbol{\kappa}_f$ produces one feature map $\mathbf{h}_f$ independently
- Output tensor: $C_{out} \times H' \times W'$
- Parameters: $C_{out} \times C_{in} \times k \times k + C_{out}$ (with bias)

Convolution as Matrix Multiplication

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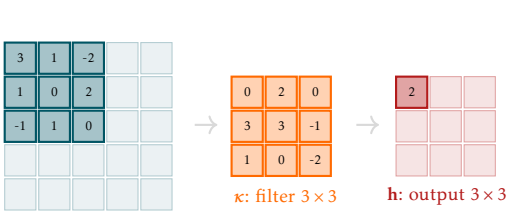
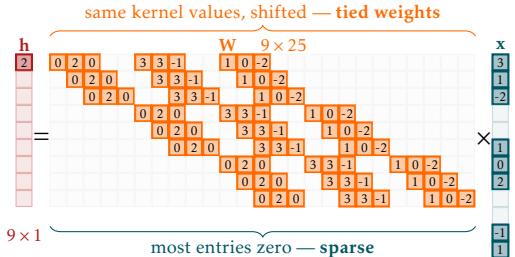
Flatten input and output — convolution becomes matrix multiplication





Convolution = FC layer with sparse and tied weights

Convolution - linear transformation!

Deep CNNs

Receptive Field

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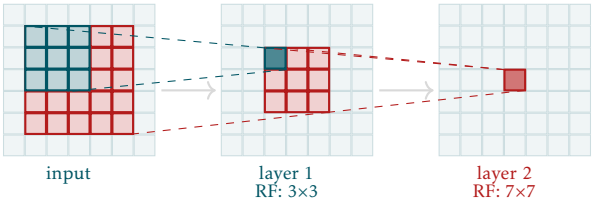
Receptive field: region of the input that influences a given neuron's output



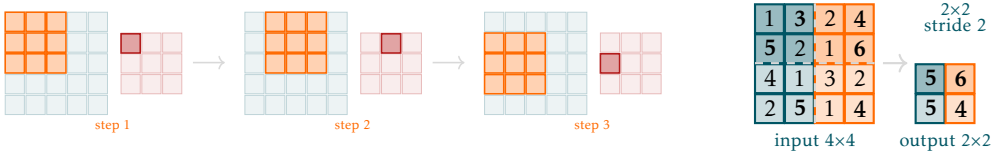
- Each 3×3 conv layer adds 2 pixels to the receptive field in each dimension
- L layers of 3×3 filters: receptive field $= (2L+1) \times (2L+1)$
- Deep networks see large regions despite small local filters
- Early layers: local features (edges) — later layers: global patterns (objects)

Pooling

Pooling

Problem: convs features spatially redundant - overlapping receptive field → nearby cells highly correlated

Solution - pooling: fixed non-trainable kernel slides through; non-overlapping regions (streid = pool size); no channel interaction (independent per channel)



Max pooling (most common):

$$y[i,j] = \max_{p,q \in \mathcal{R}_{ij}} x[p,q]$$

Average pooling:

$$y[i,j] = \frac{1}{|\mathcal{R}_{ij}|} \sum_{p,q \in \mathcal{R}_{ij}} x[p,q]$$

Retains strongest activation in each region

Effect: $C \times H \times W \xrightarrow{2\times2 \text{ pool}} C \times H/2 \times W/2$ — channel dimension unchanged

Global Average Pooling

Problem: flattening feature maps before FC layers is expensive

$C \times H \times W$ flattened $\rightarrow C \cdot H \cdot W$ inputs to FC layer — huge parameter count (VGG: ~100M parameters in FC layers alone)

Global average pooling (GAP): average each feature map to a single value

$$y_c = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c[i,j] \qquad c = 1, \ldots, C$$

$C \times H \times W \xrightarrow{\text{GAP}} C \times 1 \times 1$ — one value per feature map

Intuition: each feature map represents one concept — its average activation measures how present that concept is in the image

Benefits:

- Eliminates large FC layers — dramatic parameter reduction
- Works with any input image size — no fixed spatial dimension required

CNN Architecture

CNN Building Block

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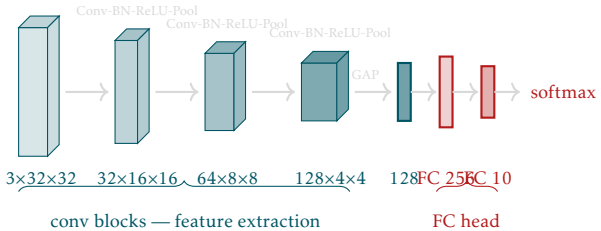
Standard pattern: Conv → BatchNorm → ReLU → (Pool)

- **Conv:** extract features via learned filters
- **BatchNorm:** stabilise activations — essential for deep CNNs
- **ReLU:** non-linearity — enables complex feature combinations
- **Pool:** spatial downsampling — reduce resolution, increase receptive field

Full CNN Architecture

Two stages: feature extraction (conv blocks) → classification (FC head)



Conv blocks:

- Increasing channels: 3 → 32 → 64 → 128
- Decreasing spatial size via pooling
- Learn hierarchical features

FC head:

- Flatten spatial features
- One or more FC layers
- Final softmax for classification

LeNet-5 (LeCun et al., 1998)

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First successful CNN — handwritten digit recognition (MNIST)

- 2 conv layers + 2 FC layers
- 28×28 input $\rightarrow$ 10 class output
- ~60k parameters — tiny by modern standards
- Used sigmoid/tanh activations, average pooling

Significance:

- Demonstrated that learned features outperform hand-crafted ones
- Introduced the conv $\rightarrow$ pool $\rightarrow$ conv $\rightarrow$ pool $\rightarrow$ FC pattern
- Deployed commercially for cheque reading

Limitation: too shallow and small for complex tasks — needed more compute

AlexNet (Krizhevsky et al., 2012)

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ImageNet moment — won ILSVRC 2012 with 15.3% top-5 error vs 26.2% runner-up

- 5 conv layers + 3 FC layers, ~60M parameters
- Key innovations:
  - ReLU activations instead of sigmoid/tanh — faster training
  - Dropout in FC layers — regularisation
  - Data augmentation — random crops, horizontal flips
  - GPU training — two GTX 580 GPUs
  - Local response normalisation (since superseded by BatchNorm)

Impact: launched the modern deep learning era — proved deep CNNs work at scale

VGG (Simonyan & Zisserman, 2014)

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Simplicity through depth — all 3×3 convolutions, increasing depth

- VGG-16: 13 conv layers + 3 FC layers, ~138M parameters
- Only 3×3 filters throughout — simplicity as a design principle
- Depth: 16–19 layers

Key insight: two 3×3 conv layers have same receptive field as one 5×5 but:

- Fewer parameters: $2 \times 3^2 = 18$ vs $5^2 = 25$
- More non-linearities — more expressive

Limitation: 3 large FC layers contain most parameters (~100M) — memory intensive

Legacy: VGG features widely used for transfer learning; clean design still instructive

ResNet (He et al., 2016)

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Problem: deeper networks were performing worse than shallower ones

Degradation problem: adding more layers hurts training accuracy — not overfitting, but optimisation failure

Key idea: **residual connections** (skip connections)

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x}$$

- Instead of learning $\mathbf{y} = \mathcal{F}(\mathbf{x})$, learn the **residual** $\mathcal{F}(\mathbf{x}) = \mathbf{y} - \mathbf{x}$
- If optimal solution is identity, easier to learn $\mathcal{F}(\mathbf{x}) = 0$ than $\mathcal{F}(\mathbf{x}) = \mathbf{x}$
- Gradient flows directly through skip connection — no vanishing gradients

Result: successfully trained networks with 50, 101, 152 layers

Residual Block

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Basic residual block:

$$\mathbf{y} = \text{ReLU}(\mathcal{F}(\mathbf{x}) + \mathbf{x})$$

$\mathcal{F}(\mathbf{x})$: two Conv-BN-ReLU layers

Bottleneck block (ResNet-50+):

- 1×1 conv: reduce channels
- 3×3 conv: spatial processing
- 1×1 conv: restore channels

Fewer parameters than basic block at same depth

Projection shortcut: when dimensions change, use 1×1 conv on skip connection to match

1×1 Convolutions

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1×1 conv: applies a learned linear combination across channels at each spatial location

$$y_c[i,j] = \sum_{c'=1}^{C_{\text{in}}} w_{c,c'}\, x_{c'}[i,j] + b_c$$

Equivalent to: FC layer applied independently at each spatial location

Uses:

- **Channel reduction/expansion** — change C_{in} to C_{out} cheaply (bottleneck blocks)
- **Non-linearity injection** — add ReLU between channels without spatial processing
- **Network-in-network** — increase model capacity without larger filters
- **Projection shortcuts** — match dimensions in residual connections

Depthwise Separable Convolutions

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Standard conv: $C_{\text{out}} \times C_{\text{in}} \times k \times k$ parameters

Depthwise separable conv: split into two steps

Depthwise conv:

- One filter per input channel
- Spatial processing only
- $C_{\text{in}} \times k \times k$ parameters

Pointwise conv (1×1)**:**

- Mix channels
- $C_{\text{out}} \times C_{\text{in}}$ parameters

Parameter reduction: factor of $\approx k^2$ — for 3×3: $\sim 9\times$ fewer parameters

Used in: MobileNet, EfficientNet — efficient architectures for mobile/edge deployment

Batch Norm in CNNs

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BatchNorm2d: normalises over batch **and** spatial dimensions for each channel

$$\mu_c = \frac{1}{N \cdot H \cdot W} \sum_{n,i,j} x_c^{(n)}[i,j], \qquad \sigma_c^2 = \frac{1}{N \cdot H \cdot W} \sum_{n,i,j} (x_c^{(n)}[i,j] - \mu_c)^2$$

- One mean and variance per **channel** — not per spatial location
- Learnable γ_c and β_c per channel

Placement: Conv → BN → ReLU (standard) or Conv → ReLU → BN (less common)

Effect in CNNs:

- Essential for training deep CNNs — without it, networks beyond ~10 layers are very hard to train
- Allows higher learning rates
- Reduces need for careful initialisation

Dropout in CNNs

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Standard dropout on FC layers works well in CNNs too

Problem with dropout on conv layers:

- Adjacent pixels are highly correlated
- Dropping individual activations does not create enough independence
- Information easily propagated through neighbouring non-dropped activations

SpatialDropout2d: drop entire feature maps (channels) rather than individual activations

- Removes all spatial locations of a channel simultaneously
- Forces network to use diverse feature maps
- More effective regularisation for conv layers

In practice: BatchNorm often sufficient regularisation in conv layers — dropout mainly used in FC head

Summary

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Why CNNs:

- Local connectivity
- Parameter sharing
- Translation invariance

Building blocks:

- Conv: feature extraction
- BatchNorm + ReLU: stability + non-linearity
- Pooling / stride: downsampling
- FC head: classification

Key architectures:

- LeNet → AlexNet → VGG → ResNet
- Residual connections: key to deep networks

Modern details:

- 1×1 conv: channel mixing
- Depthwise separable: efficiency
- BatchNorm2d: per-channel normalisation
- SpatialDropout: conv regularisation

In PyTorch:

- `nn.Conv2d`
- `nn.MaxPool2d`
- `nn.BatchNorm2d`
- `nn.AdaptiveAvgPool2d`

Next: L10 — Sequence Models & RNNs